

Markov Processes: Homework 1

Due Wed Sept 2 by 11:59pm on Gradescope

Plotting and reporting. Label all axes, state all parameters, and use consistent limits across panels. Report numerical results (empirical means, variances, probabilities) to 3–4 significant digits. Screenshots or photos of screens are not accepted.

Code. Code need not appear in the writeup. Create a public github repository named “Markov2026” and upload runnable code labeled by problem, e.g. “HW1_problem6.m”. Give your account name and repository link at the top of your submission.

Math. Accompany calculations with complete sentences and discuss what the results mean.

Optional problems. Problems after the END box, and parts marked optional, are not graded. They are worth your time anyway, and the random sums problem previews machinery we will need for Poisson processes.

1. **Joint distributions.** Random variables X, Y have joint PDF

$$f(x, y) = \begin{cases} K(x^2 + y), & 0 \leq x \leq 1, 0 \leq y \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find K .
 - (b) Find the marginal densities f_X and f_Y , and check that each integrates to one.
 - (c) Are X and Y independent?
2. **Moment generating function 1.** A basketball player shoots free throws until they make their first basket. On each attempt the probability of making the shot is $q \in (0, 1)$, and shots are independent. Let X be the number of *misses* before the first make, so $X \in \{0, 1, 2, \dots\}$.
- (a) Derive the moment generating function (MGF) of X and state where it is finite.
 - (b) Use the MGF to compute $E[X]$ and $\text{Var}(X)$.
 - (c) (*Memorylessness*) Fix an integer $k \geq 0$. Show that the MGF of $X - k$ conditional on $\{X \geq k\}$ equals $M_X(s)$, and state what this says about a player who has already missed k shots.
 - (d) Let T_r be the number of misses before the r th make. Use MGFs to identify the distribution of T_r , and give $E[T_r]$ and $\text{Var}(T_r)$.
3. **Mixtures and conditioning.** Draw $\theta \sim \text{Uniform}[0, 1]$. Conditional on θ , let $X \mid \theta \sim \text{Exponential}(\theta)$, i.e. with *rate* θ . Find the marginal PDF $f_X(x)$ and its support, justifying which θ contribute for a given x . Determine whether $E[X]$ is finite and explain.
4. **Moment generating function 2 (Exponential waiting times $\rightarrow$ Gaussian).** Let $X \sim \text{Exponential}(\lambda)$ with density $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$.
- (a) Derive the MGF of X and use it to compute $E[X]$ and $\text{Var}(X)$. State where the MGF is finite.
 - (b) Let $S_n = X_1 + \dots + X_n$ with X_i i.i.d. $\text{Exponential}(\lambda)$. Use MGFs to identify the distribution of S_n .

Optional Parts: Central Limit Theorem Extension. (Not graded. Parts (a) and (b) above are required.)

- (c) (*Standardization*) Define

$$Z_n = \frac{S_n - n/\lambda}{\sqrt{n}/\lambda} = \frac{\lambda S_n - n}{\sqrt{n}}.$$

Show that

$$M_{Z_n}(s) = e^{-s\sqrt{n}} \left(1 - \frac{s}{\sqrt{n}}\right)^{-n}, \quad s < \sqrt{n}.$$

- (d) Prove that $M_{Z_n}(s) \rightarrow e^{s^2/2}$ as $n \rightarrow \infty$ and identify the limit distribution. Comment on the size of the leading correction.

Hint: take logs and expand with $u = s/\sqrt{n}$, using $\log(1 - u) = -u - \frac{u^2}{2} - \frac{u^3}{3} + O(u^4)$.

- (e) (*Simulation*) For $n = 2^k$, $k = 1, \dots, 8$, generate 10^5 samples of Z_n , estimate the sample skewness, and plot it against n on log-log axes with the theoretical $2/\sqrt{n}$. Overlay histograms of Z_n at $n = 4$ and $n = 64$ on the standard normal density.

5. **Luck, Skill, and Upsets.** Competitors i and j have skill scores s_i and s_j , and only the difference $s = s_i - s_j$ matters. With probability $\alpha \in [0, 1]$ a match is a pure coin flip, won by either player with probability $\frac{1}{2}$. Otherwise it is decided by skill, and i wins with the logistic probability

$$\sigma(\beta s) = \frac{1}{1 + e^{-\beta s}}, \quad \beta > 0.$$

Call α the *luck* and β the *depth of competition*. This is the model of Jerdee and Newman, *Science Advances* (2024), generalizing Bradley-Terry, recovered at $\alpha = 0$, $\beta = 1$.

- (a) Show that the probability that i beats j is

$$f(s) = \frac{\alpha}{2} + (1 - \alpha) \frac{1}{1 + e^{-\beta s}}.$$

Verify $f(-s) = 1 - f(s)$, give $\lim_{s \rightarrow \pm\infty} f(s)$, and interpret the limits.

- (b) Take $\alpha = 0.2$, $\beta = 1$, and $s = 2$, so i is the stronger player. Compute $P(i \text{ wins})$ and the expected net return on a \$1 wager on i at even odds. Also find the probability the coin decided the match, *given* that j won.
- (c) Let E be “the weaker player won” and F be “the match was decided by the coin,” still with $s > 0$. Are E and F independent? Determine every (α, s) for which they are.
- (d) (*Why luck and depth are hard to separate.*) Expand f about $s = 0$ and show that

$$f(s) = \frac{1}{2} + \frac{(1 - \alpha)\beta}{4} s + O(s^3).$$

Explain why a dataset consisting only of closely matched contests cannot distinguish a large luck parameter from a small depth of competition, and describe what kind of match would break the tie.

6. **Simulation Assignment 1: Coupled Inequalities.** Let X , Y , and Z be independent, Uniform $[0, 1]$ random variables.

- (a) Analytically compute the probability that

$$X^2 + Y^2 < Z \quad \text{and} \quad Z^2 > XY.$$

Describe the region of the unit cube satisfying both inequalities and set up the integrals. (Hint: polar coordinates in x, y . The two lower bounds on z exchange dominance at some $r_*(\theta)$.)

- (b) Estimate the same probability by Monte Carlo and compare. Plot the estimate against sample size N with a logarithmic horizontal axis, and draw the analytic value as a horizontal line.

END

7. **Optional Practice: Random sums and PGFs.** (Preview of machinery we will need for Poisson processes.) Let $N, X_1, X_2, \dots$ be independent, with N and each X_i taking values in $\{0, 1, 2, \dots\}$ and the X_i identically distributed. Set

$$Y = \sum_{i=1}^N X_i, \quad (Y = 0 \text{ if } N = 0),$$

and write $G_X(t) = E[t^X]$, $G_N(t) = E[t^N]$, $G_Y(t) = E[t^Y]$.

- (a) Show that $G_Y(t) = G_N(G_X(t))$ for $|t| \leq 1$.
 - (b) Take $N \sim \text{Poisson}(\lambda)$ and $X_i \sim \text{Bernoulli}(p)$. Identify the distribution of Y and interpret the result.
 - (c) Use G_Y to show $E[Y] = E[N]E[X]$ and $\text{Var}(Y) = E[N]\text{Var}(X) + \text{Var}(N)E[X]^2$, and check these against part (b).
8. **Optional Practice: a ratio of dependent variables.** With X, Y as in Problem 1, let $V = \frac{X}{Y+1}$. Find the PDF $f_V(v)$ for all real v . (Hint: find the CDF first, and watch for the value of v at which the inner limit of integration hits the edge of the support.)
9. **Optional Practice Simulation Assignment 2: Geometric Probability with a Twist.** Let X and Y be independent random variables with PDF

$$f(u) = 2u, \quad 0 \leq u \leq 1,$$

and let $Z \sim \text{Uniform}[0, 1]$, independent of X and Y .

- (a) Show that the probability that $X^2 + Y^2 < Z$ can be written as a triple integral involving the joint PDF of (X, Y, Z) . Evaluate the integral exactly.
- (b) Estimate the probability in part (a) using Monte Carlo simulation. Generate samples from the PDF $f(u) = 2u$ for X and Y by inverse transform sampling, keep Z uniform, and compute the fraction of cases satisfying the inequality. Explain why the estimate converges to the exact value as the number of trials increases.

$$1) f(x, y) = \begin{cases} k(x^2 + y), & 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

d)

$$1 = \int_0^1 \int_0^2 k(x^2 + y) dy dx$$

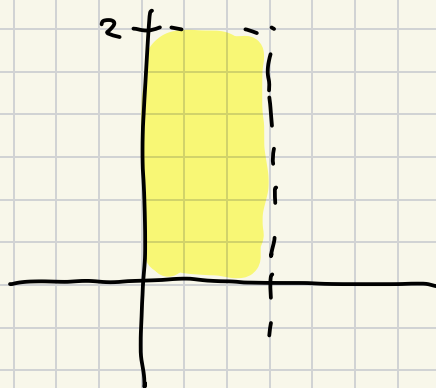
$$\frac{1}{k} = \int_0^1 \int_0^2 x^2 dy dx + \int_0^1 \int_0^2 y dy dx$$

$$= \int_0^1 2x^2 dx + \int_0^1 \left. \frac{1}{2} y^2 \right|_0^2 dx$$

$$= \left. \frac{2}{3} x^3 \right|_0^1 + \int_0^1 2 dx$$

$$= \frac{2}{3} + 2 = \frac{8}{3}$$

$$k = \frac{3}{8}$$



b) $f_x(x) = \int_0^2 \frac{3}{8} (x^2 + y) dy$

$$= \frac{3}{8} \int_0^2 x^2 + y dy$$

$$= \frac{3}{8} \left(yx^2 + \frac{1}{2} y^2 \right) \Big|_0^2$$

$$= \frac{3}{8} (2x^2 + 2)$$

$$= \left(\frac{3}{4} x^2 + \frac{3}{4} \right), \text{ when } 0 \leq x \leq 1$$

$$\int_0^1 \left(\frac{3}{4} x^2 + \frac{3}{4} \right) dx$$

$$\left. \frac{1}{4} x^3 + \frac{3}{4} x \right|_0^1$$

$$= \frac{1}{4} + \frac{3}{4} = 1 \quad \checkmark$$

$$f_y(y) = \int_0^1 \frac{3}{8} (x^2 + y) dx$$

$$= \frac{3}{8} \left(\frac{1}{3} x^3 + xy \right) \Big|_0^1$$

$$= \frac{3}{8} \left(\frac{1}{3} + y \right)$$

$$= \left(\frac{1}{8} + \frac{3}{8} y \right), \text{ when } 0 \leq y \leq 2$$

$$\int_0^2 \left(\frac{1}{8} + \frac{3}{8} y \right) dy$$

$$\left(\frac{1}{8} y + \frac{3}{16} y^2 \right) \Big|_0^2$$

$$\left(\frac{1}{4} + \frac{3}{4} \right) = 1 \quad \checkmark$$

c) $f_x(x) \cdot f_y(y) = \left(\frac{3}{4} x^2 + \frac{3}{4} \right) \left(\frac{1}{8} + \frac{3}{8} y \right)$

$$= \frac{3}{32} x^2 + \frac{9}{32} x^2 y + \frac{3}{32} + \frac{9}{32} y$$

$$= \frac{3}{8} \left(\frac{1}{4} x^2 + \frac{3}{4} x^2 y + \frac{3}{4} y + \frac{1}{4} \right) \neq \frac{3}{8} (x^2 + y)$$

Not independent

$$2) \quad X \sim \text{Geometric}(q) \quad X \in \{0, 1, 2, \dots\}$$

$$p(x) = (1-q)^x (q)$$

$$\phi_X(s) = E[e^{xs}]$$

$$= \sum_{x=0}^{\infty} e^{xs} \cdot (1-q)^x \cdot q$$

$$= q \sum_{x=0}^{\infty} e^{xs} \cdot (1-q)^x$$

$$= q \sum_{x=0}^{\infty} (e^s)^x \cdot (1-q)^x$$

$$= q \sum_{x=0}^{\infty} (e^s(1-q))^x, \text{ given } e^s(1-q) < 1$$

$$= q \frac{1}{1 - e^s(1-q)}$$

$$= \frac{q}{1 - e^s(1-q)}, \text{ finite when } e^s < \frac{1}{1-q}$$

$$b) \quad \phi'_X(s) = \frac{d}{ds} \left(\frac{q}{1 - e^s(1-q)} \right)$$

$$= q(1 - e^s(1-q))^{-1}$$

$$= -q(1 - e^s(1-q))^{-2} \cdot -(1-q)e^s$$

$$= \frac{-q \cdot -(1-q)e^s}{(1 - e^s(1-q))^2}$$

$$= \frac{q(1-q)e^s}{(1 - e^s(1-q))^2}$$

$$\phi_X(0) = \frac{q(1-q)}{(1 - (1-q))^2} = \frac{q(1-q)}{q^2} = \boxed{\frac{1-q}{q}} = E[X]$$

$$\phi''_X(s) = \frac{(1 - e^s(1-q))^2 (q(1-q)e^s) - (q(1-q)e^s) \cdot 2 \cdot (1 - e^s(1-q)) \cdot -(1-q)e^s}{(1 - e^s(1-q))^4}$$

$$\phi''_X(0) = \frac{(1 - 1 + q)^2 (q(1-q)) - (q(1-q)) \cdot 2 \cdot q \cdot (-1 + q)}{q^4}$$

$$\begin{aligned}
 2) b) \phi_X''(0) &= \frac{(1-q+q)^2(q(1-q)) - (q(1-q)) \cdot 2 \cdot q \cdot (1-q)}{q^4} \\
 &= \frac{q^3(1-q) + 2q^2(1-q)^2}{q^4} \\
 &= \frac{q(1-q) + 2(1-q)^2}{q^2} \\
 &= \frac{(1-q)(q + 2(1-q))}{q^2}
 \end{aligned}$$

$$\begin{aligned}
 \text{var}(X) &= \frac{(1-q)(q + 2(1-q))}{q^2} - \left(\frac{1-q}{q}\right)^2 \\
 &= \frac{(1-q)(q + 2(1-q)) - (1-q)^2}{q^2} \\
 &= \frac{(1-q)(q + 2(1-q) - (1-q))}{q^2} \\
 &= \boxed{\frac{1-q}{q^2}}
 \end{aligned}$$

$$\begin{aligned}
 * c) \phi_{X-k|X \geq k}(s) &= E[e^{s(X-k)} | X \geq k] \quad \text{b/c } E[g(X)] = \sum g(x) \cdot P(X) \\
 &= \sum_{n=0}^{\infty} e^{s(n-k)} P(X=n | X \geq k) \\
 &= \sum_{n=k}^{\infty} e^{s(n-k)} \cdot \frac{P(X=n \cap X \geq k)}{P(X \geq k)} \\
 &= \sum_{n=k}^{\infty} e^{s(n-k)} \cdot \frac{P(X=n)}{(1-q)^k} \quad \leftarrow \text{since } X \text{ will always be } \geq k \text{ in this summation} \\
 &= \sum_{n=k}^{\infty} e^{s(n-k)} \cdot \frac{(1-q)^n \cdot q}{(1-q)^k}
 \end{aligned}$$

$$2) c) \phi_{X-K|X \leq K}(s) = \underbrace{q e^{-sK}}_{(1-q)^K} \sum_{h=1}^{\infty} (e^s)^h \cdot (1-q)^h$$

$$= \frac{q e^{-sK}}{(1-q)^K} \cdot \frac{(e^s(1-q))^K}{1 - e^s(1-q)}$$

$$= \frac{q}{1 - e^s(1-q)} = \phi_X(s)$$

$$\sum_{n=1}^{\infty} r^n = r^1 + r^{1+1} + r^{1+1+1} + \dots$$

$$\sum_{n=0}^{\infty} r^n = 1 + r^1 + r^2 + \dots$$

d) $T_r = \sum_{i=1}^r X_i$ where X_i are i.i.d r.v.s w/ same dist as X

$$\phi_{T_r}(s) = \phi_{\sum_{i=1}^r X_i}(s) = (\phi_{X_i}(s))^r$$

$$= \left(\frac{q}{1 - e^s(1-q)} \right)^r$$

$$\phi'_{T_r}(s) = r \left(q(1 - e^s(1-q))^{-1} \right)^{r-1} \cdot -q(1 - e^s(1-q))^{-2} \cdot - (1-q)e^s$$

$$\phi'_{T_r}(s) = r \left(q(1 - e^s(1-q))^{-1} \right)^{r-1} \cdot \frac{q(1-q)e^s}{(1 - e^s(1-q))^2}$$

$$\phi'_{T_r}(0) = r \left(q(1 - (1-q))^{-1} \right)^{r-1} \cdot \frac{1-q}{q}$$

$$E[T_r] = \boxed{\frac{r(1-q)}{q}}$$

$$\phi''_{T_r}(s) = r \left(q(1 - e^s(1-q))^{-1} \right)^{r-1} \cdot \frac{(1-q)^2(q(1-q)) - (q(1-q))^2 \cdot 2 \cdot q \cdot (-1+q)}{q^4}$$

$$+ \frac{q(1-q)e^s}{(1 - e^s(1-q))^2} \cdot r(r-1) \left(q(1 - e^s(1-q))^{-1} \right)^{r-2} \cdot \frac{q(1-q)e^s}{(1 - e^s(1-q))^2}$$

$$\phi''_{T_r}(0) = r \cdot \frac{(1-q)(q+2(1-q))}{q^2} + \frac{1-q}{q} \cdot \frac{r(r-1)(1-q)}{q}$$

$$= \frac{r(1-q)(q+2(1-q)) + r(r-1)(1-q)^2}{q^2}$$

$$\text{Var}(T_r) = \frac{r(1-q)(q+2(1-q)) + r(r-1)(1-q)^2}{q^2} - \left(\frac{r(1-q)}{q} \right)^2$$

z) d)

$$\text{Var}(T_r) = \frac{r(1-q)(q+2(1-q)) + r(r-1)(1-q)^2}{q^2} - \frac{r^2(1-q)^2}{q^2}$$

$$= \frac{r(1-q)(-q+2 + (r-1)(1-q)) - r(1-q)}{q^2}$$

$$= \frac{r(1-q)(-q+2 + r - rq - 1 + q - r + rq)}{q^2}$$

$$= \frac{r(1-q)(-1)}{q^2} = \frac{-r(1-q)}{q^2}$$

$$3) \theta \sim \text{Uniform}(0,1) \quad X|\theta \sim \text{Exponential}(\theta)$$

$$f_{X|\theta}(x, \theta) = \frac{f_{X,\theta}(x, \theta)}{f_{\theta}(\theta)}$$

$$f_{X,\theta}(x, \theta) = f_{X|\theta}(x|\theta) \cdot f_{\theta}(\theta)$$

$$f_{X,\theta}(x, \theta) = \theta e^{-\theta x} \cdot 1$$

$$f_{X,\theta}(x, \theta) = \theta e^{-\theta x} \quad \text{where } 0 < \theta \leq 1 \text{ and } x > 0$$

$$f_X(x) = \int_0^1 \theta e^{-\theta x} d\theta \quad \begin{array}{l} u = \theta \quad v' = e^{-\theta x} \\ u' = 1 \quad v = -\frac{1}{x} e^{-\theta x} \end{array} \quad \begin{array}{l} \frac{d}{dx}(uv) = u v' + u' v \\ uv = \int u v' dx + \int u' v dx \end{array}$$

$$= \left[-\frac{\theta}{x} e^{-\theta x} - \int -\frac{1}{x} e^{-\theta x} d\theta \right]_0^1 \quad \int u v' dx = uv - \int u' v dx$$

$$= \left(-\frac{\theta}{x} e^{-\theta x} - \frac{1}{x^2} e^{-\theta x} \right)_0^1$$

$$= \left(-\frac{1}{x} e^{-x} - \frac{1}{x^2} e^{-x} \right) - \left(0 - \frac{1}{x^2} \right)$$

$$= \left(-e^{-x} \left(\frac{1}{x} + \frac{1}{x^2} \right) + \frac{1}{x^2} \right), \text{ where } x > 0$$

any $0 < \theta \leq 1$ contributes to any given x

$$E[X] = \int_0^{\infty} x \cdot \left(-e^{-x} \left(\frac{1}{x} + \frac{1}{x^2} \right) + \frac{1}{x^2} \right) dx$$

$$E[X] = E[E[X|\theta]]$$

$$= E\left[\frac{1}{\theta}\right]$$

$$= \infty$$

$$= \int_0^{\infty} -e^{-x} - \frac{1}{x} e^{-x} + \frac{1}{x} dx$$

$$= \int_0^{\infty} -e^{-x} dx - \int_0^{\infty} \frac{e^{-x} - 1}{x} dx \quad \begin{array}{l} \text{approaches} \\ \frac{1-1}{x} \text{ at origin} \end{array}$$

$\Rightarrow \frac{1}{x}$ diverges (harmonic series)

Because as $x \rightarrow \infty$, $e^{-x} \rightarrow 0$, the integrand approaches $-\frac{1}{x}$. Since the harmonic series diverges, $E[X]$ is infinite.

$$4) X \sim \text{Exponential}(\lambda)$$

$$f(x) = \lambda e^{-\lambda x}, x \geq 0$$

$$d) \phi_X(s) = E[e^{sx}]$$

$$= \int_0^{\infty} e^{sx} \cdot \lambda e^{-\lambda x} dx$$

$$= \lambda \int_0^{\infty} e^{x(s-\lambda)} dx$$

Assuming $s < \lambda$

$$= \lambda \left(\frac{1}{s-\lambda} e^{x(s-\lambda)} \right) \Big|_0^{\infty}$$

$$= \lambda \left(0 - \frac{1}{s-\lambda} \right)$$

$$\phi_X(s) = \frac{-\lambda}{s-\lambda}, \text{ finite when } s < \lambda$$

$$\phi'_X(s) = \frac{\lambda}{(s-\lambda)^2} \quad \phi'_X(0) = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda} = E[X]$$

$$\phi''_X(s) = \frac{-2\lambda}{(s-\lambda)^3} \quad \phi''_X(0) = \frac{2}{\lambda^2} \quad \text{var}(\lambda) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

$$5) S_n = X_1 + \dots + X_n \quad X_i \sim \text{Exponential}(\lambda)$$

$$\phi_{S_n}(s) = \left(\frac{-\lambda}{s-\lambda} \right)^n$$

$$\phi'_{S_n}(s) = n \left(\frac{-\lambda}{s-\lambda} \right)^{n-1} \cdot \frac{\lambda}{(s-\lambda)^2}$$

$$\phi'_{S_n}(0) = n(1)^{n-1} \cdot \frac{1}{\lambda} = \frac{n}{\lambda} = E[S_n]$$

$$\phi''_{S_n}(s) = n(n-1) \left(\frac{-\lambda}{s-\lambda} \right)^{n-2} \cdot \frac{\lambda}{(s-\lambda)^2} \cdot \frac{\lambda}{(s-\lambda)^2} + n \left(\frac{-\lambda}{s-\lambda} \right)^{n-1} \cdot \frac{-2\lambda}{(s-\lambda)^3}$$

$$\phi''_{S_n}(0) = n(n-1) \cdot \frac{1}{\lambda^2} + n \cdot \frac{2}{\lambda^2} = \frac{n(n-1)+2n}{\lambda^2} = \frac{n(n+1)}{\lambda^2}$$

$$\text{var}(S_n) = \frac{n(n+1)}{\lambda^2} - \frac{n^2}{\lambda^2} = \frac{n}{\lambda^2} - \text{gamma distribution}$$

$$s) \quad \sigma(\beta s) = \frac{1}{1+e^{-\beta s}}, \quad \beta > 0$$

$$a) \quad f(s) = \alpha \left(\frac{1}{2} \right) + (1-\alpha) \sigma(\beta s) \\ = \frac{\alpha}{2} + (1-\alpha) \left(\frac{1}{1+e^{-\beta s}} \right)$$

$$f(-s) = \frac{\alpha}{2} + (1-\alpha) \left(\frac{1}{1+e^{\beta s}} \right) \cdot \frac{e^{-\beta s}}{e^{-\beta s}} \\ = \frac{\alpha}{2} + (1-\alpha) \left(\frac{e^{-\beta s}}{1+e^{-\beta s}} \right) \\ = \frac{\alpha}{2} + (1-\alpha) \left(\frac{1+e^{-\beta s}}{1+e^{-\beta s}} - \frac{1}{1+e^{-\beta s}} \right) \\ = \frac{\alpha}{2} + (1-\alpha) - (1-\alpha) \left(\frac{1}{1+e^{-\beta s}} \right) \\ = 1 - \frac{\alpha}{2} - (1-\alpha) \left(\frac{1}{1+e^{-\beta s}} \right) \\ = 1 - f(s)$$

$$\lim_{s \rightarrow \infty} \left(\frac{\alpha}{2} + (1-\alpha) \left(\frac{1}{1+e^{-\beta s}} \right) \right)$$

$$= \frac{\alpha}{2} + (1-\alpha) = 1 - \frac{\alpha}{2}$$

$$\lim_{s \rightarrow -\infty} \left(\frac{\alpha}{2} + (1-\alpha) \left(\frac{1}{1+e^{-\beta s}} \right) \right)$$

$$= \frac{\alpha}{2}$$

As player i gets infinitely better than player j , the chance he wins approaches 1 - (the chance the game is determined by luck: chance he wins a 50/50 coin flip)

As player i gets infinitely worse the chance he wins is the chance the game is determined by luck: $\frac{1}{2}$

$$5) b) f(2) = \frac{0.2}{2} + (1-0.2)\left(\frac{1}{1+e^{-2}}\right)$$

$$P(i \text{ wins}) = \frac{1}{10} + \frac{4}{5}\left(\frac{1}{1+e^{-2}}\right) \approx 0.8046$$

$$w(x) = \begin{cases} 2 & \text{if } x \\ 0 & \text{otherwise} \end{cases}$$

$$E[W] = 2 \cdot 0.8046 + 0 \cdot (1-0.8046)$$

≈ 1.609 , Expected net return ≈ 0.609 per game

Prob that coin decided the match $= 0.2$

Prob that j won $= 0.1954$

$$P(C | j) = \frac{P(j|C) \cdot P(C)}{P(j)} \quad \text{— Bayes' theorem}$$

$$= \frac{0.5 \cdot 0.2}{0.1954} \approx \boxed{0.5119}$$

c) E = "weaker person won"
 F = "match was decided by coin"

$$E = 1 - f(s), \text{ assuming } s > 0$$

$$= \frac{\alpha}{2} + (1-\alpha)\frac{1}{1+e^{\beta s}}$$

$$F = \alpha$$

$$P(E | F) = \frac{1}{2}$$

so if α and s are chosen such that $E = \frac{1}{2}$, then E and F are indep

$$\frac{1}{2} = \frac{\alpha}{2} + (1-\alpha)\frac{1}{1+e^{\beta s}}$$

$$0 = \frac{\alpha-1}{2} + (1-\alpha)\frac{1}{1+e^{\beta s}}$$

$$0 = (1+e^{\beta s})(\alpha-1) + 2(1-\alpha)$$

$$= \alpha + \alpha e^{\beta s} - 1 - e^{\beta s} + 2 - 2\alpha$$

$$0 = -\alpha + \alpha e^{\beta s} - e^{\beta s} + 1$$

$$0 = (\alpha-1)(e^{\beta s} - 1)$$

} indep when $\alpha=1$ or when $s=0$,
 but since $s > 0$, we only take
 $\alpha=1$

5)d) Maclaurin series: $f(x) = \sum_{i=0}^{\infty} \frac{f^{(i)}(0) \cdot x^i}{i!}$

$$f(s) = f(0) + f'(0) \cdot s + \frac{f''(0)s^2}{2} + \frac{f'''(0)s^3}{6}$$

$$f(0) = \frac{1}{2}$$

$$f'(s) = -(1-\alpha) \cdot \frac{1}{(1+e^{-\beta s})^2} \cdot -\beta e^{-\beta s}$$

$$= (1-\alpha) \cdot \frac{\beta e^{-\beta s}}{(1+e^{-\beta s})^2}$$

$$f'(0) = (1-\alpha) \cdot \frac{\beta}{4}$$

$$f''(s) = \frac{(1+e^{-\beta s})^2 \cdot (-\beta^2 e^{-\beta s}) - \beta e^{-\beta s} \cdot 2(1+e^{-\beta s}) \cdot (-\beta e^{-\beta s})}{(1+e^{-\beta s})^4} \cdot (1-\alpha)$$

$$= \frac{-\beta^2 e^{-\beta s} (1+e^{-\beta s})^2 + 2\beta^2 e^{-2\beta s} \cdot (1+e^{-\beta s})}{(1+e^{-\beta s})^4} \cdot (1-\alpha)$$

$$= \frac{-\beta^2 e^{-\beta s} ((1+e^{-\beta s}) - 2e^{-\beta s})}{(1+e^{-\beta s})^3} \cdot (1-\alpha)$$

$$= (1-\alpha) \cdot \frac{\beta^2 e^{-\beta s} (e^{-\beta s} - 1)}{(1+e^{-\beta s})^3}$$

$$f''(0) = 0$$

$$f(s) = \frac{1}{2} + (1-\alpha) \frac{\beta}{4} \cdot s + 0 + \dots$$

As $s \rightarrow 0$, the terms containing α also $\rightarrow 0$. The major term remaining, then, is the constant $\frac{1}{2}$. This is why a luck factor, α , doesn't play a large role across a large number of evenly matched games.

As s increases or decreases, the luck factor α becomes more influential, eventually being the only parameter that matters as $s \rightarrow \pm\infty$.

$$6) X, Y, Z \sim \text{Uniform}(0, 1)$$

$$a) X^2 + Y^2 < Z$$

$$\sqrt{XY} < Z$$

using cylindrical coords,

$$r^2 < Z$$

$$\sqrt{r^2 \cos \theta \sin \theta} < Z$$

$$X^2 + Y^2 = \sqrt{XY}$$

$$r^2 = r \sqrt{\cos \theta \sin \theta}$$

$$r = \sqrt{\cos \theta \sin \theta}$$

$$r = \frac{\sqrt{\sin(2\theta)}}{\sqrt{2}}$$

! b/c there's no upper bound for Z, r finds $\max(E_1, E_2)$

$$\begin{aligned} P &= \int_0^{\frac{\pi}{2}} \int_0^{\frac{\sqrt{\sin(2\theta)}}{\sqrt{2}}} \int_0^1 r^2 \, dz \, dr \, d\theta + \int_0^{\frac{\pi}{2}} \int_0^{\frac{\sqrt{\sin(2\theta)}}{\sqrt{2}}} \int_{\frac{\sqrt{\sin(2\theta)}}{\sqrt{2}}}^1 r \, dz \, dr \, d\theta \\ &= \int_0^{\frac{\pi}{2}} \int_0^{\frac{\sqrt{\sin(2\theta)}}{\sqrt{2}}} (r - r^3) \, dr \, d\theta + \int_0^{\frac{\pi}{2}} \int_0^{\frac{\sqrt{\sin(2\theta)}}{\sqrt{2}}} (r - r^2 \frac{\sqrt{\sin(2\theta)}}{\sqrt{2}}) \, dr \, d\theta \\ &= \int_0^{\frac{\pi}{2}} \left(\frac{1}{2} - \frac{1}{4} \right) \left(\frac{\sin(2\theta)}{4} - \frac{\sin^2(2\theta)}{16} \right) d\theta + \int_0^{\frac{\pi}{2}} \left(\frac{\sin(2\theta)}{4} - \frac{\sin^2(2\theta)}{12} \right) d\theta \\ &= \int_0^{\frac{\pi}{2}} \frac{\left(\frac{1 - \cos(4\theta)}{2} \right) - 4\sin(2\theta) + 4}{16} d\theta + \int_0^{\frac{\pi}{2}} \frac{3\sin(2\theta) - \left(\frac{1 - \cos(4\theta)}{2} \right)}{12} d\theta \\ &= \int_0^{\frac{\pi}{2}} \frac{-\cos(4\theta) - 8\sin(2\theta) + 9}{32} d\theta + \int_0^{\frac{\pi}{2}} \frac{6\sin(2\theta) - 1 + \cos(4\theta)}{24} d\theta \\ &= \left(\frac{\sin(4\theta)}{128} + \frac{\cos(2\theta)}{8} + \frac{9\theta}{32} \right) \Big|_0^{\frac{\pi}{2}} + \left(\frac{-\cos(2\theta)}{8} - \frac{\theta}{24} + \frac{\sin(4\theta)}{96} \right) \Big|_0^{\frac{\pi}{2}} \\ &= 0 + \frac{(-1)}{8} + \frac{9\pi}{64} - \frac{1}{8} + \frac{1}{8} - \frac{\pi}{48} + \frac{1}{8} \\ &= \frac{9\pi}{64} - \frac{\pi}{48} = \frac{23\pi}{192} \end{aligned}$$